

```
import keras
from keras.models import Sequential
from keras.layers import Dense

#Create Sequential model with Dense Layers, using the add method
model = Sequential()

#Dense implements the operation:
#    output = activation(dot(input, kernel) + bias)
#Units are the dimensionality of the output space for the layer,
#    which equals the number of hidden units
#Activation and Loss functions may be specified by strings or classes
model.add(Dense(units=64, activation='relu', input_dim=100))
model.add(Dense(units=10, activation='softmax'))
```

Building a sequential model in Keras

that re-use layers. It is not very flexible to allow for a combination of models using layers like Merge, Concatenate and Add. In a sequential model, we stack layers sequentially on top of one another. Hence, each layer has unique input and output, which have unique input and output shapes. We can retrieve all those weights that a layer has as a list of numpy arrays. Each layer has its defining configuration that you can get by calling *get_config* on a layer.

Models with Functional API

The functional API enables you to define more complex models, such as directed acyclic graphs, multi-output models, or models with shared layers. To create a model with the functional API we need to compose a set of input, output layers, and then pass them to the `keras_model` function. Since you can connect layers to any other layer, therefore complex models can be created easily.

The Keras functional API provides a more flexible way of defining models. Specifically, it allows you to define multiple outputs or input models as well as models that share layers. The Functional API was designed to make it easier to build neural network models that are non-sequential, as most of the Neural Networks research was pointing in that direction.

The old Graph API could do that but it was very difficult to use, as you had to name each node and refer to it by name. The Functional API makes it easy to build complex models. It can be used to create flexible model architecture since each layer can be connected in a pairwise fashion. It can create complex networks such as Siamese networks and residual networks